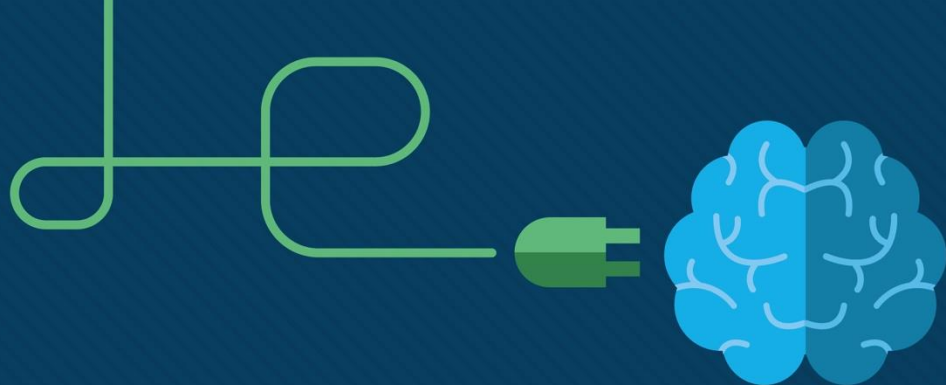




Module 6: EtherChannel

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: EtherChannel

Module Objective: Troubleshoot EtherChannel on switched links.

Topic Title	Topic Objective
EtherChannel Operation	Describe EtherChannel technology.
Configure EtherChannel	Configure EtherChannel.
Verify and Troubleshoot EtherChannel	Troubleshoot EtherChannel.

6.1 EtherChannel Operation

EtherChannel Operation

Link Aggregation

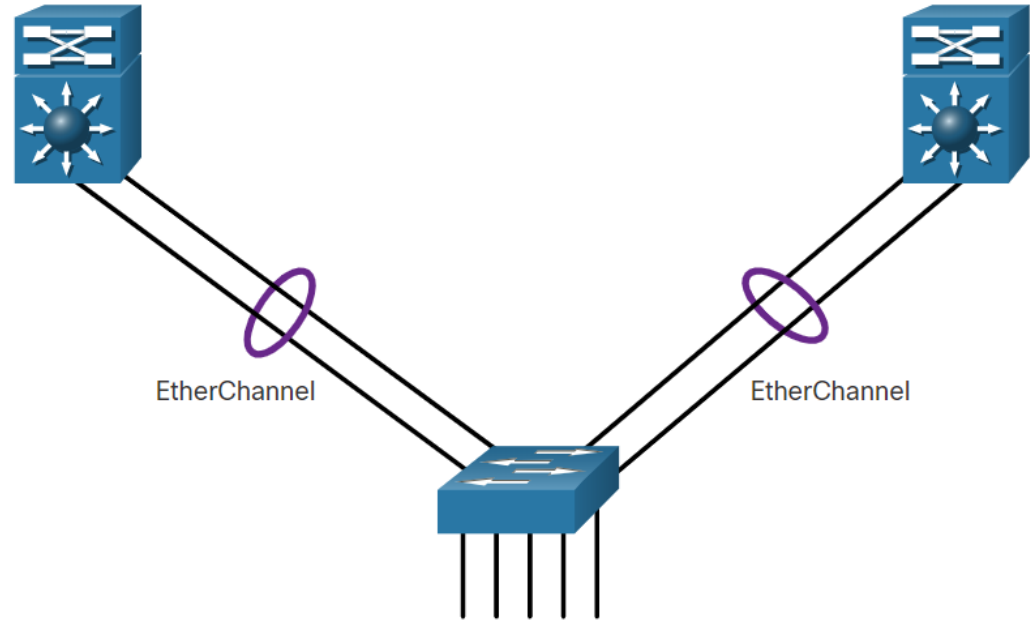
- There are scenarios in which **more bandwidth** or **redundancy** between devices is needed than what can be provided by a single link. **Multiple links** could be connected between devices to increase bandwidth. However, Spanning Tree Protocol (**STP**), which is enabled on Layer 2 devices like Cisco switches by default, will **block redundant links** to prevent switching **loops**.
- A **link aggregation** technology is needed that allows redundant links between devices that will not be blocked by STP. That technology is known as **EtherChannel**.
- EtherChannel is a link aggregation technology that **groups multiple physical** Ethernet links together into one **single logical link**. It is used to provide **fault-tolerance, load sharing, increased bandwidth**, and **redundancy** between switches, routers, and servers.
- EtherChannel technology makes it possible to **combine** the **number of physical links** between the switches to increase the overall speed of switch-to-switch communication.

EtherChannel Operation

EtherChannel

EtherChannel technology was originally developed by Cisco as a LAN switch-to-switch technique of grouping several Fast Ethernet or Gigabit Ethernet ports into one logical channel.

When an EtherChannel is configured, the resulting **virtual interface** is called a **port channel**. The physical interfaces are **bundled** together into a port channel interface, as shown in the figure.



Advantages of EtherChannel

EtherChannel technology has many **advantages**, including the following:

- Most **configuration** tasks can be done on the **EtherChannel** interface instead of on each individual port, ensuring configuration consistency throughout the links.
- EtherChannel relies on existing switch ports. There is no need to upgrade the link to a faster and more expensive connection to have **more bandwidth**.
- **Load balancing** takes place between links that are part of the same EtherChannel.
- EtherChannel creates an **aggregation** that is seen as **one logical link**. When several EtherChannel bundles exist between two switches, STP may block one of the bundles to prevent switching loops. When STP blocks one of the redundant links, it blocks the entire EtherChannel. This blocks all the ports belonging to that EtherChannel link. Where there is only one EtherChannel link, all physical links in the EtherChannel are active because **STP** sees only one (logical) link.
- EtherChannel provides redundancy because the overall link is seen as one logical connection. Additionally, the **loss of one physical link** within the channel does **not create a change** in the topology.

EtherChannel Operation

Implementation Restrictions

EtherChannel has certain implementation **restrictions**, including the following:

- **Interface types cannot be mixed**. For example, Fast Ethernet and Gigabit Ethernet cannot be mixed within a single EtherChannel.
- Currently each EtherChannel can consist of **up to eight** compatibly-configured Ethernet **ports**. EtherChannel provides full-duplex bandwidth up to 800 Mbps (Fast EtherChannel) or 8 Gbps (Gigabit EtherChannel) between one switch and another switch or host.
- The Cisco Catalyst **2960** Layer 2 switch currently supports **up to six EtherChannels**.
- The individual EtherChannel group member port configuration must be consistent on both devices. If the physical ports of one side are configured as trunks, the physical ports of the other side must also be configured as trunks within the **same native VLAN**. Additionally, all ports in each EtherChannel link must be configured as Layer 2 ports.
- Each EtherChannel has a logical **port channel interface**. A configuration applied to the port channel interface affects all physical interfaces that are assigned to that interface.

EtherChannel Operation

AutoNegotiation Protocols

EtherChannels can be formed through **negotiation** using one of two protocols, **Port Aggregation Protocol (PAgP)** or **Link Aggregation Control Protocol (LACP)**. These protocols allow ports with similar characteristics to form a channel through dynamic negotiation with adjoining switches.

Note: It is also possible to configure a **static or unconditional EtherChannel** without PAgP or LACP.

EtherChannel Operation

PAgP Operation

PAgP (pronounced “Pag - P”) is a **Cisco-proprietary protocol** that aids in the automatic creation of EtherChannel links. When an EtherChannel link is configured using PAgP, PAgP packets are sent between EtherChannel-capable ports to **negotiate** the forming of a channel. When PAgP identifies matched Ethernet links, it groups the links into an EtherChannel. The EtherChannel is then added to the spanning tree as a single port.

When enabled, PAgP also manages the EtherChannel. **PAgP packets** are sent every **30 seconds**. PAgP checks for **configuration consistency** and manages **link additions and failures** between two switches. It ensures that when an EtherChannel is created, all ports have the same type of configuration.

Note: In EtherChannel, it is mandatory that all ports have the **same speed, duplex setting, and VLAN** information. Any port modification after the creation of the channel also changes all other channel ports.

EtherChannel Operation

PAgP Operation (Cont.)

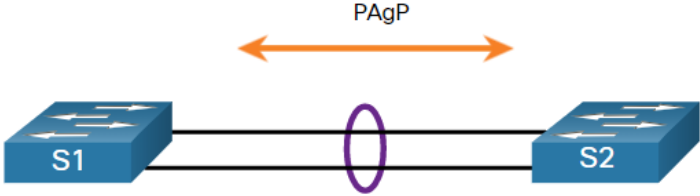
PAgP helps create the EtherChannel link by detecting the configuration of each side and ensuring that links are compatible so that the EtherChannel link can be enabled when needed. The **modes for PAgP** as follows:

- **On** - This mode forces the interface to **channel without PAgP**. Interfaces configured in the on mode do not exchange PAgP packets.
- **PAgP desirable** - This PAgP mode places an interface in an **active negotiating state** in which the interface initiates negotiations with other interfaces by **sending PAgP packets**.
- **PAgP auto** - This PAgP mode places an interface in a **passive negotiating state** in which the interface **responds** to the PAgP packets that it receives but **does not initiate PAgP negotiation**.

The **modes must be compatible** on each side. If one side is configured to be in auto mode, it is placed in a passive state, waiting for the other side to initiate the EtherChannel negotiation. If the other side is also set to auto, the negotiation never starts and the EtherChannel does not form. If all modes are disabled by using the **no** command, or if no mode is configured, then the EtherChannel is disabled. The on mode manually places the interface in an EtherChannel, without any negotiation. It works only if the other side is also set to on. If the other side is set to negotiate parameters through PAgP, no EtherChannel forms, because the side that is set to **on mode does not negotiate**. No negotiation between the two switches means there is no checking to make sure that all the links in the EtherChannel are terminating on the other side, or that there is PAgP compatibility on the other switch.

EtherChannel Operation

PAgP Mode Settings Example



The table shows the various combination of PAgP modes on S1 and S2 and the resulting channel establishment outcome.

S1	S2	Channel Establishment
On	On	Yes
On	Desirable/Auto	No
Desirable	Desirable	Yes
Desirable	Auto	Yes
Auto	Desirable	Yes
Auto	Auto	No

EtherChannel Operation

LACP Operation

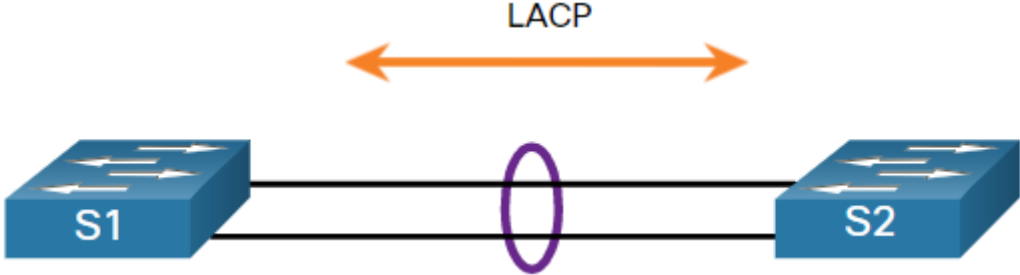
LACP is part of an **IEEE specification (802.3ad)** that allows several **physical ports** to be **bundled** to form a **single logical channel**. LACP allows a switch to negotiate an automatic bundle by sending **LACP packets** to the other switch. It performs a function similar to PAgP with Cisco EtherChannel. Because LACP is an IEEE standard, it can be used to facilitate EtherChannels in **multivendor** environments. On Cisco devices, both protocols are supported.

LACP provides the same negotiation benefits as PAgP. LACP helps create the EtherChannel link by detecting the configuration of each side and making sure that they are compatible so that the EtherChannel link can be enabled when needed. The **modes for LACP** are as follows:

- **On** - This mode forces the interface to **channel without LACP**. Interfaces configured in the on mode do **not exchange LACP packets**.
- **LACP active** - This LACP mode places a port in an **active negotiating** state. In this state, the port initiates negotiations with other ports by sending LACP packets.
- **LACP passive** - This LACP mode places a port in a **passive negotiating** state. In this state, the port **responds** to the LACP packets that it receives but does **not initiate** LACP packet negotiation.

EtherChannel Operation

LACP Mode Settings Example



The table shows the various combination of LACP modes on S1 and S2 and the resulting channel establishment outcome.

S1	S2	Channel Establishment
On	On	Yes
On	Active/Passive	No
Active	Active	Yes
Active	Passive	Yes
Passive	Active	Yes
Passive	Passive	No

6.2 Configure EtherChannel

Configure EtherChannel

Configuration Guidelines

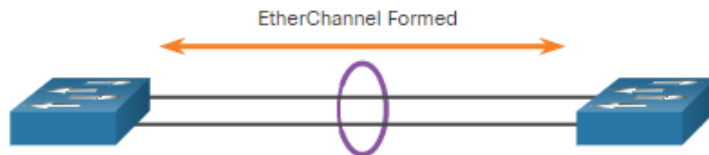
The following guidelines and restrictions are useful for configuring EtherChannel:

- **EtherChannel support** - All Ethernet interfaces must support EtherChannel with no requirement that interfaces be physically contiguous.
- **Speed and duplex** - Configure all interfaces in an EtherChannel to operate at the **same** speed and in the same duplex mode.
- **VLAN match** - All interfaces in the EtherChannel bundle must be assigned to the **same** VLAN or be configured as a trunk (shown in the figure).
- **Range of VLANs** - An EtherChannel supports the **same allowed range** of VLANs on all the interfaces in a trunking EtherChannel. If the allowed range of VLANs is not the same, the interfaces do not form an EtherChannel, even when they are set to **auto** or **desirable** mode.

Configure EtherChannel

Configuration Guidelines (Cont.)

- The figure shows a configuration that would allow an EtherChannel to form between S1 and S2.
- If these settings must be changed, **configure** them **in port channel interface configuration** mode. Any configuration that is applied to the port channel interface also **affects individual interfaces**. However, configurations that are applied to the individual interfaces do not affect the port channel interface. Therefore, making configuration changes to an interface that is part of an EtherChannel link may cause interface compatibility **issues**.
- The **port channel** can be configured in **access mode**, **trunk mode** (most common), or on a **routed port**.



S1 Port Configurations

Speed	1 Gbps
Duplex	Full
VLAN	10

S2 Port Configurations

Speed	1 Gbps
Duplex	Full
VLAN	10

Configure EtherChannel

LACP Configuration Example

Configuring EtherChannel with LACP requires the following three steps:

- **Step 1.** Specify the interfaces that compose the EtherChannel group using the **interface range interface** global configuration mode command. The **range** keyword allows you to select several interfaces and configure them all together.
- **Step 2.** Create the port channel interface with the **channel-group identifier mode active** command in interface range configuration mode. The identifier specifies a **channel group number**. The **mode active** keywords identify this as an **LACP** EtherChannel configuration.
- **Step 3.** To change Layer 2 settings on the port channel interface, enter port channel interface configuration mode using the **interface port-channel** command, followed by the interface identifier. In the example, S1 is configured with an LACP EtherChannel. The port channel is configured as a trunk interface with the allowed VLANs specified.

```
S1(config)# interface range FastEthernet 0/1 - 2
S1(config-if-range)# channel-group 1 mode active
Creating a port-channel interface Port-channel 1
S1(config-if-range)# exit
S1(config-if)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 1,2,20
```

Packet Tracer – Configure EtherChannel

In this Packet Tracer, you will complete the following objectives:

- Configure Basic Switch Settings
- Configure an EtherChannel with Cisco PAgP
- Configure and 802.3ad EtherChannel
- Configure a Redundant EtherChannel Link

6.3 Verify and Troubleshoot EtherChannel

Verify and Troubleshoot EtherChannel

Verify EtherChannel

As always, when you configure devices in your network, you must verify your configuration. If there are problems, you will also need to be able to troubleshoot and fix them. There are a number of commands to verify an EtherChannel configuration:

- The **show interfaces port-channel** command displays the general status of the port channel interface.
- The **show etherchannel summary** command displays one line of information per port channel.
- The **show etherchannel port-channel** command displays information about a specific port channel interface.
- The **show interfaces etherchannel** command can provide information about the role of a physical member interface of the EtherChannel.

Common Issues with EtherChannel Configurations

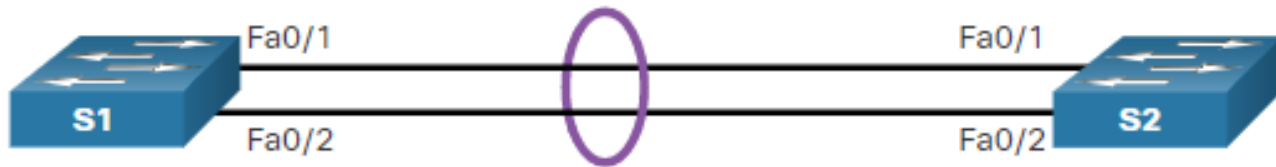
All interfaces within an EtherChannel must have the **same** configuration of **speed** and **duplex** mode, **native** and **allowed VLANs** on trunks, and **access VLAN** on access ports. Ensuring these configurations will significantly reduce network problems related to EtherChannel. Common EtherChannel issues include the following:

- Assigned **ports** in the EtherChannel are not part of the **same VLAN**, or not configured as trunks. Ports with different native VLANs cannot form an EtherChannel.
- Trunking was configured on some of the ports that make up the EtherChannel, but not all of them. It is not recommended that you configure trunking mode on individual ports that make up the EtherChannel. When **configuring** a trunk on an EtherChannel, verify the trunking mode on the **EtherChannel**.
- If the allowed **range of VLANs** is not the same, the ports do not form an EtherChannel even when PAgP is set to the **auto** or **desirable** mode.
- The dynamic **negotiation** options for **PAgP** and **LACP** are **not compatibly** configured on both ends of the EtherChannel.

Verify and Troubleshoot EtherChannel

Troubleshoot EtherChannel Example

In the figure, interfaces F0/1 and F0/2 on switches S1 and S2 are connected with an EtherChannel. However, the EtherChannel is not operational.



Verify and Troubleshoot EtherChannel

Troubleshoot EtherChannel Example (Cont.)

Step 1. View the EtherChannel Summary Information: The output of the **show etherchannel summary** command indicates that the EtherChannel is down.

```
S1# show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        N - not in use, no aggregation
       f - failed to allocate aggregator
       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
       A - formed by Auto LAG
Number of channel-groups in use: 1
Number of aggregators:          1
Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SD)        -           Fa0/1(D)  Fa0/2(D)
```

Troubleshoot EtherChannel Example (Cont.)

Step 2. View Port Channel Configuration: In the **show run | begin interface port-channel** output, more detailed output indicates that there are incompatible PAgP modes configured on S1 and S2.

```
S1# show run | begin interface port-channel
interface Port-channel1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
!
interface FastEthernet0/1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode on
!
interface FastEthernet0/2
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode on
!
=====
S2# show run | begin interface port-channel
interface Port-channel1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
!
interface FastEthernet0/1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode desirable
!
interface FastEthernet0/2
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode desirable
```

Troubleshoot EtherChannel Example (Cont.)

Step 3: Correct the Misconfiguration: To correct the issue, the PAgP mode on the EtherChannel is changed to desirable.

Note: EtherChannel and STP must interoperate. For this reason, the order in which EtherChannel-related commands are entered is important, which is why you see interface Port-Channel 1 removed and then re-added with the **channel-group** command, as opposed to directly changed. If one tries to change the configuration directly, STP errors cause the associated ports to go into blocking or errdisabled state.

```
S1(config)# no interface port-channel 1
S1(config)# interface range fa0/1 - 2
S1(config-if-range)# channel-group 1 mode desirable
Creating a port-channel interface Port-channel 1
S1(config-if-range)# no shutdown
S1(config-if-range)# exit
S1(config)# interface range fa0/1 - 2
S1(config-if-range)# channel-group 1 mode desirable
S1(config-if-range)# no shutdown
S1(config-if-range)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# end
S1#
```

Troubleshoot EtherChannel Example (Cont.)

Step 4. Verify EtherChannel is Operational: The EtherChannel is now active as verified by the output of the **show etherchannel summary** command.

```
S1# show etherchannel summary
Flags: D - down          P - bundled in port-channel
      I - stand-alone    s - suspended
      H - Hot-standby (LACP only)
      R - Layer3         S - Layer2
      U - in use        N - not in use, no aggregation
      f - failed to allocate aggregator
      M - not in use, minimum links not met
      m - not in use, port not aggregated due to minimum links not met
      u - unsuitable for bundling
      w - waiting to be aggregated
      d - default port
      A - formed by Auto LAG
Number of channel-groups in use: 1
Number of aggregators:          1
Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        PAgP       Fa0/1(P)  Fa0/2(P)
```

Packet Tracer – Troubleshoot EtherChannel

In this Packet Tracer, you will complete the following:

- Examine the Physical Layer and Correct Switch Port Mode Issues
- Identify and Correct Port Channel Assignment Issues
- Identify and Correct Port Channel Assignment Issues

6.4 Module Practice and Quiz

Packet Tracer – Implement EtherChannel

In this Packet Tracer, you will complete the following:

- Build the Network
- Configure EtherChannel

Lab – Implement EtherChannel

In this lab, you will complete the following objectives:

- Part 1: Build the Network and Configure Basic Device Settings
- Part 2: Create VLANs and Assign Switch Ports
- Part 3: Configure 802.1Q Trunks between the Switches
- Part 4: Implement and Verify an EtherChannel between the switches

What Did I Learn In This Module?

- To increase **bandwidth** or **redundancy**, multiple links could be connected between devices. However, **STP** will block redundant links to prevent switching loops. **EtherChannel** is a **link aggregation technology** that allows redundant links between devices that will not be blocked by STP.
- EtherChannel groups **multiple physical Ethernet links** together into one **single logical link**. It provides **fault-tolerance**, **load sharing**, **increased bandwidth**, and **redundancy** between switches, routers, and servers.
- When an EtherChannel is configured, the resulting **virtual interface** is called a **port channel**.
- EtherChannels can be formed through **negotiation** using one of two protocols, **PAgP** or **LACP**. These protocols allow ports with similar characteristics to form a channel through dynamic negotiation with adjoining switches.
- When an EtherChannel link is configured using **Cisco-proprietary PAgP**, PAgP packets are sent between EtherChannel-capable ports to negotiate the forming of a channel. Modes for PAgP are **On**, **PAgP desirable**, and **PAgP auto**.
- LACP performs a function similar to PAgP with Cisco EtherChannel. Because **LACP is an IEEE** standard, it can be used to facilitate EtherChannels in **multivendor** environments. Modes for LACP are **On**, **LACP active**, and **LACP passive**.

What Did I Learn In This Module? (Cont.)

- The following **guidelines and restrictions** are useful for configuring EtherChannel:
 - All Ethernet **interfaces** on all modules must **support EtherChannel** with no requirement that interfaces be physically contiguous, or on the same module.
 - Configure all interfaces in an EtherChannel to operate at the **same speed** and in the same **duplex** mode.
 - All interfaces in the EtherChannel bundle must be assigned to the **same VLAN** or be configured as a trunk.
 - An EtherChannel supports the **same allowed range of VLANs** on all the interfaces in a trunking EtherChannel.
- Configuring EtherChannel with LACP requires three steps:
 - Step 1. Specify the interfaces that compose the EtherChannel group using the **interface range interface** global configuration mode command.
 - Step 2. Create the port channel interface with the **channel-group identifier mode active** command in interface range configuration mode.
 - Step 3. To change Layer 2 settings on the port channel interface, enter port channel interface configuration mode using the **interface port-channel** command, followed by the interface identifier.

What Did I Learn In This Module? (Cont.)

- There are a number of commands to verify an EtherChannel configuration including **show interfaces port-channel**, **show etherchannel summary**, **show etherchannel port-channel**, and **show interfaces etherchannel**.
- Common EtherChannel issues include the following:
 - Assigned **ports** in the EtherChannel are **not part of the same VLAN**, or not configured as trunks.
 - Ports with **different native VLANs** cannot form an EtherChannel.
 - Trunking was configured on **some of the ports** that make up the EtherChannel, but not all of them.
 - If the allowed range of **VLANs is not the same**, the ports do not form an EtherChannel even when PAgP is set to the auto or desirable mode.
 - The dynamic **negotiation options** for PAgP and LACP are **not compatibly** configured on both ends of the EtherChannel.

New Terms and Commands

- Link Aggregation
- EtherChannel
- Port Channel
- Port Aggregation Protocol (PAgP)
- Link Aggregation Control Protocol (LACP)
- PAgP desirable
- PAgP auto
- LACP active
- LACP passive
- **channel-group X mode [desirable | auto | active | passive]**
- **interface port-channel X**
- **show interfaces port-channel**
- **show etherchannel summary**
- **show etherchannel port-channel**
- **show interfaces etherchannel**

